

Data Card — African Language Wikipedia Corpus for SuperBPE Tokenization

Project: Bridging the Tokenization Gap in African LLMs via SuperBPE **Program:** TRI AI Saturdays **Status:** 8 of 8 target languages complete (Swahili, Chichewa, Zulu, Kinyarwanda, Hausa, Amharic, Yoruba, Wolof — per Section 7 of the project plan), plus Igbo as a documented bonus 9th language beyond the required set. Wolof required a second, supplementary source beyond Wikipedia to reach an adequate corpus size; this is documented in Sections 2.1 and 6, consistent with the collection plan's risk mitigation for uneven corpus size across languages.

1. Motivation

Standard BPE tokenizers, trained mostly on English-dominated web text, over-fragment morphologically rich and agglutinative African languages, inflating token counts and degrading downstream LLM performance. This corpus was collected to train and evaluate SuperBPE (Liu et al., 2025, arXiv:2503.13423) tokenizers on real African-language text, to measure whether "superword" tokenization meaningfully reduces fragmentation and improves efficiency across language families with different morphological structures (Bantu agglutination, Volta-Niger and West Atlantic tonal marking, Chadic concatenative morphology, Semitic root-and-pattern morphology with a non-Latin script).

2. Composition

Language	Family	Script	Source	Raw extracted	Final cleaned	Retention
Swahili (sw)	Bantu	Latin	Wikipedia	150,426	86,866	57.7%
Zulu (zu)	Bantu	Latin	Wikipedia	14,273	3,859	27.0%
Yoruba (yo)	Volta-Niger	Latin	Wikipedia	47,996	13,833	28.8%

Language	Family	Script	Source	Raw extracted	Final cleaned	Retention
Igbo (ig) <i>(bonus)</i>	Volta-Niger	Latin	Wikipedia	57,457	53,686	93.4%
Hausa (ha)	Chadic	Latin	Wikipedia	114,318	104,631	91.5%
Chichewa (ny)	Bantu	Latin	Wikipedia	1,408	892	63.4%
Amharic (am)	Semitic	Ethiopic	Wikipedia	23,189	8,522	36.8%
Kinyarwanda (rw)	Bantu	Latin	Wikipedia	13,850	11,125	80.3%
Wolof (wo) †	West Atlantic	Latin	Wikipedia + MasakhaNER	2,351	3,721	see 2.1
Total				425,268	286,135	mixed sources — see per-row

All Wikipedia-sourced languages draw from dumps.wikimedia.org monthly snapshots, licensed [cc-by-sa-4.0](https://creativecommons.org/licenses/by-sa/4.0/) (text only). Retention rate varies substantially by language — this reflects genuine differences in how each language's Wikipedia is composed (see Section 6), not inconsistent cleaning.

† Wolof's "Raw extracted" and "Retention" columns reflect the Wikipedia portion only. "Final cleaned" (3,721) is the combined total after adding a supplementary source — see breakdown below.

2.1 Wolof multi-source breakdown

Wolof Wikipedia alone (1,073 documents, 405,389 words after cleaning) fell well short of every other language in this corpus and was flagged during spot-check as likely insufficient before cleaning even confirmed it, consistent with the collection plan's instruction to flag insufficient data before drawing conclusions. It was supplemented with MasakhaNER 1.0's Wolof split (Adelani et al., 2021; github.com/masakhane-io/masakhane-ner), a human-annotated NER dataset sourced from local news articles, licensed [CC-BY-4.0-NC](https://creativecommons.org/licenses/by-nc/4.0/). NER tags were discarded and sentences reconstructed as plain text; train/dev/test splits were combined since the original task split is irrelevant once the data is used as raw tokenizer training text rather than labeled NER examples.

Source	Domain	Documents/Sentences	Final words
Wikipedia	Encyclopedic	1,073	405,389
MasakhaNER 1.0	News (human-annotated)	2,648	52,777
Combined		3,721	458,166

Even combined, Wolof's total word count (458,166) remains well below Kinyarwanda's (1,931,619) and every other language except Chichewa — documented as a genuine, unresolved shortfall rather than papered over by the second source.

3. Collection Process

Each language followed an identical procedure, so cross-language comparisons reflect real linguistic/data differences rather than uneven handling. Wolof is the one documented exception (Section 2.1):

1. Register the source (license, access method, expected coverage) in the project's Source Registry before any data was pulled.
2. Download the full Wikipedia dump; extract clean article text via `wikiextractor`, explicitly excluding non-article namespaces (categories, talk pages, user pages).
3. Manually spot-check a random sample of 20 extracted articles per language (two independent samples were drawn for Swahili, Zulu, and Yoruba) against three criteria: language correctness, encoding integrity, and diacritic/spelling intactness.
4. Apply an 8-rule automated cleaning pipeline (Section 4) informed directly by what the manual spot-checks found, and re-applied retroactively to earlier languages whenever a new rule was added.
5. **Wolof only:** after cleaning confirmed the Wikipedia corpus was substantially smaller than every other language's, source and convert a supplementary corpus (MasakhaNER 1.0) following a parallel but distinct procedure suited to its CoNLL-tagged format (Section 2.1), then combine both sources into one corpus before the holdout split.
6. Hold out a 5% evaluation split per language, stored in a separate directory from training data (`eval_holdout/`), with a recorded document count and SHA-256 content hash to verify it is never touched during training. For Wolof, the eval split was drawn from the combined (Wikipedia + MasakhaNER) corpus, not the Wikipedia-only corpus, so train and eval reflect the same source/domain mix.

4. Cleaning Rules Applied

Rules 1–8 below were applied identically across all Wikipedia-sourced languages, including Kinyarwanda and the Wikipedia portion of Wolof:

1. **Minimum word count filter** (<15 words dropped) — addresses empty infobox/list-only page extractions, found in every language's spot-check sample. Severity varies substantially by language (10% of Igbo/Hausa's sample vs. 50% of Amharic's vs. 60% of Wolof's — see Section 6).
2. **Encoding sanity check** — drops documents matching known mojibake patterns. Zero triggered for Kinyarwanda; zero for Wolof.
3. **Exact deduplication** — content-hash based, catches verbatim repeats.
4. **Near-duplicate template capping** — masks numbers/proper nouns into a "template skeleton"; keeps at most 3 examples of any repeated skeleton. Motivated by Yoruba's spot-check (10 of 20 sampled articles were near-identical bot-generated asteroid-catalog stubs undetected by exact-hash dedup). Verified removal: 198 documents in Yoruba, 323 in Amharic, 86 in Kinyarwanda (21 distinct templates), 7 in Wolof (4 distinct templates — including a newly observed pattern of near-identical stub articles about unrelated Lithuanian football clubs, a different template family than Yoruba's astronomy stubs but the same underlying risk).
5. **Outlier-length truncation** (99th percentile cap) — mirrors the mitigation used in the original SuperBPE paper after they found a single document duplicated 2,224 times distorted their tokenizer training.
6. **Leaked XML revision-metadata removal** — found in Zulu's spot-check: raw, HTML-entity-encoded Wikipedia revision metadata leaking into extracted text via a `wikiextractor` edge case. Frequency varied hugely by language: 1-2 instances in Zulu/Chichewa, 11 in Kinyarwanda, 0 in Wolof, but **505 instances in Amharic** — a genuinely large gap worth flagging, though its underlying cause is not established (see Section 6). Fix verified correct at every observed frequency.
7. **Leaked templatestyles/infobox markup removal** — found in Igbo's spot-check: two structural variants (a clean tag-pair form, and a messier raw JSON "data-mw" form with no reliable closing boundary). The messier variant is a documented best-effort fix: it reliably removes the JSON debris but may occasionally remove a few adjacent real words with no whitespace separator (affects an estimated ~5/57,457 Igbo documents, ~0.01%). 50 instances found in Kinyarwanda; 1 in Wolof.
8. **Citation marker removal** — strips leftover `[1]`, `[2]``[3]` footnote references not fully removed by extraction. 168 instances in Kinyarwanda; 11 in Wolof.

Wolof's MasakhaNER portion used a separate, narrower procedure (not rules 1–8): a minimum sentence length of 3 words (vs. 15 for Wikipedia articles, since these are single news sentences rather than full articles) and exact deduplication only. Rules 2, 6, and 7 (encoding sanity, XML/templatestyles leaks) are Wikipedia-extraction-specific artifacts and do not apply to this source, so they were not run against it.

5. Evaluation Split

Language	Train docs	Eval docs	Eval %
Swahili	82,523	4,343	5.0%
Zulu	3,667	192	5.0%
Yoruba	13,142	691	5.0%
Igbo	51,002	2,684	5.0%
Hausa	99,400	5,231	5.0%
Chichewa	848	44	4.9%
Amharic	8,096	426	5.0%
Kinyarwanda	10,569	556	5.0%
Wolof (combined)	3,535	186	5.0%

Each split uses a fixed seed (42) for reproducibility, and each eval set has a recorded SHA-256 content hash (in its `eval_manifest.md`) so anyone can verify a given corpus file was never touched during training, per the project's requirement that eval data be held out before anything else touches the data.

6. Known Limitations

- **Empty-page extraction rates vary hugely by language and are a real property of each source Wikipedia, not a pipeline defect.** Ranges from ~5% (Kinyarwanda) and ~7% (Igbo, Hausa) to **50% (Amharic)** and **60% (Wolof)** in manually-verified 20-article samples. Amharic's failures were disproportionately bare year-number articles (e.g. "229 እ.ኤ.አ.", "1938") — infobox-only date pages with zero prose. Wolof's failures were a mix of true empty stubs (35% of its sample) plus several near-threshold short stubs, without Amharic's specific date-page pattern.
- **Wolof required a supplementary source and remains the smallest corpus by word count even after combining sources** (458,166 words, vs. Chichewa's 204,640 and every other language's 1M+). Documented per the collection plan's instruction to report shortfalls rather than pad or conceal them.
- **Chichewa Wikipedia is extremely small:** only 1,408 raw articles, 204,640 words after cleaning — well short of this project's 500,000-token working target. Documented honestly per the collection plan's instruction to report shortfalls rather than pad them.

- **Tone/diacritic marking is inconsistently applied within the Yoruba and Igbo corpora.** Confirmed directly across independent samples: the same word appears both with and without diacritics in different articles (e.g. Igbo "amụrụ" vs. "amuru" for "born"). Cannot be corrected automatically without a real morphological analyzer or native-speaker review, per the project's morphological rule toolkit consultation protocol (Section B.3). Hausa, Amharic, Kinyarwanda, and Wolof showed no equivalent pattern in their own orthographic systems in spot-check samples.
- **Leaked XML revision-metadata frequency is unexplained and highly uneven:** 505 instances in Amharic vs. 0–11 in every other language. The fix is verified correct regardless of frequency, but the cause of this concentration in Amharic specifically has not been investigated further.
- **A small number of Zulu articles (1 confirmed instance) contained informal/off-register content** (a "Polandball" internet-meme article ending in a stray "omg" fragment). Not a technical defect; noted for register/source_type tracking.
- **One unresolved, low-confidence signal remains in the Igbo corpus:** a single possible leftover citation-marker instance (1/53,686 documents, ~0.002%) that could not be located via direct text search across multiple verification attempts. Documented rather than pursued further.
- **One Kinyarwanda article in the spot-check sample showed an apparent mid-article code-switch into untranslated English** (a sentence that trails off into English text partway through). Likely a genuine Wikipedia editorial artifact (an unfinished translation) rather than an extraction bug; not currently addressed by an automated rule since it was observed once in a 20-document sample and not confirmed as a systemic pattern at full-corpus scale.

7. Cross-Language Observations

- **Non-Latin script (Amharic/Ethiopic) introduced no encoding defects.** Zero mojibake, zero wrong-language leaks across the full spot-check sample — the pipeline's encoding handling generalized correctly to a completely different writing system with no code changes required. The same holds for Kinyarwanda and Wolof's Latin-script diacritics (ë, à, ñ, η, apostrophe-marked forms), which rendered cleanly with zero encoding issues in both languages' spot-checks.
- **Retention rate is not correlated with corpus size.** Hausa (114,318 raw) and Igbo (57,457 raw) both retained >90%, while Zulu (14,273 raw) and Yoruba (47,996 raw) retained under 30%, and Wolof (2,351 raw, the smallest Wikipedia source in this corpus) retained under 50% — article count alone does not predict data quality or usability.

8. Maintenance

This corpus was assembled by the Data Engineering role for this project. Sources, cleaning code, and spot-check reports are version-controlled alongside this Data Card ([TRI-Project/](#) on the team's shared Drive). All cleaning rules were applied identically and retroactively across every Wikipedia-sourced language whenever a new rule was introduced, so cross-language comparisons in this document reflect real data differences, not inconsistent handling introduced partway through the project. Wolof's supplementary-source procedure (Section 2.1) is the one documented departure from this identical-pipeline approach, taken in direct response to a corpus-size shortfall rather than as a default step for every language.